pumped into the filter under a pressure of 200 to 700 kN/m^2. After filtration, water comes out under high pressure. The rate of filtration is 6000 to 15000 litres/m^2/hour. They are of horizontal or vertical type. They are more suitable for industries, private estates, etc. They are less efficient than slow and rapid sand filters. Inspection windows are provided at the top. A horizontal type of filter is shown in fig.

Working

The coagulated water under pressure is directly admitted to the pressure filter through inlet valve. The filtered water comes out through the outlet valve. It is collected in the central drain and conveyed to filtered water storage tank. During normal working, only the valves for raw water and filtered water are kept open.

The cleaning of the filter may be carried out by backwashing like rapid sand filter. The compressed air may be used to agitate sand grains. For cleaning, inlet and outlet valves are closed. The wash water valve and wash water gutter valve are opened. The cleaning is done more frequently.

4.5 Disinfection of Water

Disinfection of water is the process of removal of pathogenic bacteria from water by chemical or other means.

The chemicals used for killing disease producing bacterias are known as disinfectants.

4.5.1 Necessity of Disinfection

The filtered water may contain some harmful impurities such as disease producing bacterias, dissolved inorganic salts, colour, odour, taste, iron and manganese. The bacterially contaminated water will spread various diseases and their epidemics causing disaster to public life. Hence, disinfection is most essential. Further disinfection not only kills the existing bacterias from water, but also prevents its contamination during its transit from the treatment plant to place for its consumption.

4.5.2 Methods of Disinfection

There are two methods of disinfection of water. They are:

(i) Chlorination

(ii) Minor methods

(i) Chlorination

Chlorine has been universally recognized as the most ideal disinfectant in treatment of water on a large scale.

The treatment by chlorination is cheap and reliable. It produces desired effects and lasts long. It is also easy to measure and handle. Chlorination is the treatment of water with chlorine or its compounds for the disinfection of water.

Chlorination not only disinfects, but also removes colour, odour, unpleasant taste and prevents the growth of weeds in water.

(ii) Minor methods

1. Boiling
2. Excess lime treatment
3. Iodine and bromine treatment
4. Ozone treatment
5. Potassium permanganate treatment
6. Silver treatment
7. Ultra-violet rays treatment

The most commonly adopted two minor methods of disinfection of water are discussed below.

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